

Chapter-4

Eigenvalues and Eigenvectors

Let $A = (a_{ij})_{n \times n}$ be a square matrix. A non-zero vector $\mathbf{V}$ in $\mathbb{R}^n$ is called an eigenvector of A if $A\mathbf{V}$ is a scalar multiple of $\mathbf{V}$; that is $A\mathbf{V} = \lambda\mathbf{V}$ for some scalar λ . The scalar λ is called an eigenvalue of A and $\mathbf{V}$ is called the eigenvector of A corresponding to λ .

Example: The vector $\mathbf{V} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is an eigenvector of $A = \begin{pmatrix} 3 & 0 \\ 8 & -1 \end{pmatrix}$ corresponding to the eigenvalue $\lambda = 3$.

$$A\mathbf{V} = \begin{pmatrix} 3 & 0 \\ 8 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \lambda\mathbf{V}.$$

Characteristic matrix:

Provided that A is a square matrix of order $n \times n$. Then the matrix $A - \lambda I$ is called the characteristic matrix where λ is a scalar and I is an unit matrix.

Example:

$$A = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$$

Characteristic matrix is

$$A - \lambda I = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{pmatrix}$$

Characteristic polynomial:

The determinant of characteristic matrix $(A - \lambda I)$ is a polynomial of λ . Characteristic polynomial is

$$\begin{aligned} |A - \lambda I| &= \begin{vmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{vmatrix} \\ &= (2-\lambda)(6-5\lambda+\lambda^2) - 2(1-\lambda) + 1(2-3+\lambda) \\ &= \lambda^3 - 7\lambda^2 + 11\lambda - 5. \end{aligned}$$

Characteristic equation:

The equation $|A - \lambda I| = 0$ is called characteristic equation for matrix A .

For example, $\lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$ is characteristic equation for the above matrix A .

Characteristic root(s) or eigenvalue(s):

The roots of the characteristic equation $|A - \lambda I| = 0$ are called characteristic values or eigenvalues of matrix A .

$$\therefore \lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$$

$$\Rightarrow (\lambda - 1)(\lambda - 1)(\lambda - 5) = 0 \Rightarrow \lambda = 1, 1, 5$$

So, the characteristic values or eigenvalues are 1, 1 and 5.

Example:

Find the eigenvalues and eigenvectors of the matrix $A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & -2 & 0 \\ 0 & 5 & 2 \end{pmatrix}$

Solution:

The characteristic matrix of A is,

$$A - \lambda I = \begin{pmatrix} 1 & 2 & -1 \\ 0 & -2 & 0 \\ 0 & 5 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1-\lambda & 2 & -1 \\ 0 & -2-\lambda & 0 \\ 0 & 5 & 2-\lambda \end{pmatrix}$$

$$\text{The characteristic polynomial of matrix } A \text{ is } |A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 & -1 \\ 0 & -2-\lambda & 0 \\ 0 & 5 & 2-\lambda \end{vmatrix}$$

The characteristic equation of matrix A is $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 2 & -1 \\ 0 & -2-\lambda & 0 \\ 0 & 5 & 2-\lambda \end{vmatrix} = 0$$

$$\therefore (\lambda - 1)(\lambda + 2)(\lambda - 2) = 0$$

So, the characteristic roots or the eigenvalues of matrix A is $\lambda = 1, -2, 2$

Now consider $\mathbf{V} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ is an eigenvector of matrix A corresponding to the eigenvalue λ . From the definition of eigenvalue and eigenvector

$$(A - \lambda I)\mathbf{V} = 0$$

$$\begin{pmatrix} 1-\lambda & 2 & -1 \\ 0 & -2-\lambda & 0 \\ 0 & 5 & 2-\lambda \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

For $\lambda = 1$,

Augmented matrix

$$\left(\begin{array}{ccc|c} 0 & 2 & -1 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 5 & 1 & 0 \end{array} \right)$$

$$r_2 \rightarrow -\frac{1}{3}r_2 \quad \sim \quad \left(\begin{array}{ccc|c} 0 & 2 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 5 & 1 & 0 \end{array} \right)$$

$$r_3 \rightarrow r_3 - 5r_2 \quad \sim \quad \left(\begin{array}{ccc|c} 0 & 2 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right)$$

For $\lambda = 1$, system of equations becomes,

$$2v_2 - v_3 = 0$$

$$v_2 = 0$$

$$v_3 = 0$$

Solving we get $v_2 = v_3 = 0$

MatLab command for finding eigenvalues and eigenvectors:

```
>> A=[1 2 -1;0 -2 0;0 5 2];
```

```
>> [v,d]=eig(sym(A));
```

```
>> eigenvalues=eig(A)'
```

```
eigenvalues =
```

```
1 2 -2
```

```
>> eigenvectors=double(v)
```

Here, v_1 is a free variable. Let, $v_1 = a$, where a ($a \neq 0$) is any real number. Therefore, the eigenvector of A corresponding to the eigenvalue $\lambda = 1$ is the non-zero vector of the form $\mathbf{V}_1 = \begin{pmatrix} a \\ 0 \\ 0 \end{pmatrix}$.

Again, for $\lambda = -2$,

$$\left(\begin{array}{ccc|c} 3 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 5 & 4 & 0 \end{array} \right) r_2 \leftrightarrow r_3 \sim \left(\begin{array}{ccc|c} 3 & 2 & -1 & 0 \\ 0 & 5 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

So, System of linear equations becomes,

$$3v_1 + 2v_2 - v_3 = 0$$

$$5v_2 + 4v_3 = 0$$

This system has one free variable. v_3 is a free variable and let $v_3 = b$, where b ($b \neq 0$) is any real number.

$$\therefore v_2 = -\frac{4b}{5} \text{ and } \therefore v_1 = \frac{13b}{15}$$

$$\text{Therefore, } \mathbf{V}_2 = \begin{pmatrix} \frac{13b}{15} \\ -\frac{4b}{5} \\ b \end{pmatrix}.$$

For $\lambda = 2$,

$$\left(\begin{array}{ccc|c} -1 & 2 & -1 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 5 & 0 & 0 \end{array} \right) r_3 \rightarrow 4r_3 + 5r_2 \sim \left(\begin{array}{ccc|c} -1 & 2 & -1 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

So, system of linear equations becomes,

$$-v_1 + 2v_2 - v_3 = 0$$

$$-4v_2 = 0$$

Hence, $v_2 = 0$ and v_3 is free variable. Let $v_3 = c$, where c ($c \neq 0$) is any real number.

Then we have $v_1 = -c$

$$\text{Therefore, } \mathbf{V}_3 = \begin{pmatrix} -c \\ 0 \\ c \end{pmatrix}$$

Therefore, eigenvectors are $\begin{pmatrix} a \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} \frac{13b}{15} \\ -\frac{4b}{5} \\ b \end{pmatrix}$ and $\begin{pmatrix} -c \\ 0 \\ c \end{pmatrix}$ corresponding to the eigenvalues $\lambda = 1, -2, 2$ respectively.

Example: Solve the following system of differential equation using eigenvalues and eigenvectors.

$$\begin{cases} \dot{x}_1(t) = -1.5x_1(t) + 0.5x_2(t) \\ \dot{x}_2(t) = x_1(t) - x_2(t) \end{cases} \text{ with } x_1(0) = 5 \text{ and } x_2(0) = 4,$$

$$\text{where } \dot{x}_1(t) = \frac{dx_1}{dt} \text{ and } \dot{x}_2(t) = \frac{dx_2}{dt}.$$

Solution:

Let, $X(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$ and $\dot{X}(t) = \begin{pmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{pmatrix}$

So, $X(0) = \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$.

Now the system of differential equation can be written as

$\dot{X}(t) = AX(t)$, where A is the coefficient matrix and $A = \begin{pmatrix} -1.5 & 0.5 \\ 1 & -1 \end{pmatrix}$.

Let λ and $\mathbf{V}$ be the eigenvalue and eigenvector of A and C is an integral constant then we have the solution of the form,

$X(t) = C\mathbf{V}e^{\lambda t}$.

The characteristic matrix of A is $(A - \lambda I) = \begin{pmatrix} -1.5 & 0.5 \\ 1 & -1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
 $= \begin{pmatrix} -1.5 - \lambda & 0.5 \\ 1 & -1 - \lambda \end{pmatrix}$

The characteristic polynomial of A is $|A - \lambda I| = \begin{vmatrix} -1.5 - \lambda & 0.5 \\ 1 & -1 - \lambda \end{vmatrix}$

The characteristic equation of A is $|A - \lambda I| = 0$

$$\begin{aligned} &\Rightarrow \begin{vmatrix} -1.5 - \lambda & 0.5 \\ 1 & -1 - \lambda \end{vmatrix} \\ &\Rightarrow (\lambda + 1.5)(\lambda + 1) - 0.5 = 0 \\ &\Rightarrow (\lambda^2 + 2.5\lambda + 1) = 0 \\ &\Rightarrow (\lambda + 0.5)(\lambda + 2) = 0 \end{aligned}$$

So the characteristic roots or the eigenvalues of A are $\lambda = -0.5, -2$

Now, consider $\mathbf{V} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ is the eigenvector of A corresponding to the eigenvalue λ . From the definition of eigenvalue and eigenvector, $A\mathbf{V} = \lambda\mathbf{V}$

So, $(A - \lambda I)\mathbf{V} = 0$

Augmented matrix form

$$\left(\begin{array}{cc|c} -1.5 - \lambda & 0.5 & 0 \\ 1 & -1 - \lambda & 0 \end{array} \right)$$

For $\lambda = -0.5$,

$$\left(\begin{array}{cc|c} -1 & 0.5 & 0 \\ 1 & -0.5 & 0 \end{array} \right)$$

$$r_2 \rightarrow r_2 + r_1 \quad \sim \left(\begin{array}{cc|c} -1 & 0.5 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

Solving the above system, we get $-v_1 + 0.5v_2 = 0$.

Here v_2 is a free variable. Let $v_2 = a$. $v_1 = 0.5a$

Therefore, the eigenvector of A corresponding to the eigenvalue $\lambda = -0.5$ are the non-zero vectors of the form $\mathbf{V}_1 = \begin{pmatrix} 0.5a \\ a \end{pmatrix}$.

Again, for $\lambda = -2$,

$$\begin{array}{l} \text{Augmented matrix} \\ \left(\begin{array}{cc|c} 0.5 & 0.5 & 0 \\ 1 & 1 & 0 \end{array} \right) \\ r_2 \rightarrow r_2 - 2r_1 \quad \sim \quad \left(\begin{array}{cc|c} 0.5 & 0.5 & 0 \\ 0 & 0 & 0 \end{array} \right) \\ r_1 \rightarrow \frac{1}{0.5}r_1 \quad \sim \quad \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right) \end{array}$$

Solving the above system, we get $v_1 + v_2 = 0$.

Here v_2 is a free variable. Let $v_2 = b, v_1 = -b$.

Therefore, the eigenvector of A corresponding to the eigenvalue $\lambda = -2$ are the non-zero vectors of the form $\mathbf{V}_2 = \begin{pmatrix} -b \\ b \end{pmatrix}$.

So, the solution of the system of differential equation can be written as,

$$\begin{aligned} X(t) &= C_1 \mathbf{V}_1 e^{\lambda_1 t} + C_2 \mathbf{V}_2 e^{\lambda_2 t} \\ \Rightarrow \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} &= C_1 \begin{pmatrix} 0.5a \\ a \end{pmatrix} e^{-0.5t} + C_2 \begin{pmatrix} -b \\ b \end{pmatrix} e^{-2t} \\ \Rightarrow \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} &= aC_1 \begin{pmatrix} 0.5 \\ 1 \end{pmatrix} + bC_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} \\ \Rightarrow \begin{pmatrix} 5 \\ 4 \end{pmatrix} &= aC_1 \begin{pmatrix} 0.5 \\ 1 \end{pmatrix} + bC_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} \end{aligned}$$

We can write,

$$0.5aC_1 - bC_2 = 5$$

$$aC_1 + bC_2 = 4$$

Solving the system for C_1 and C_2 , we have, $C_1 = \frac{6}{a}$ and $C_2 = \frac{-2}{b}$. Now,

$$\begin{aligned} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} &= 6 \begin{pmatrix} 0.5 \\ 1 \end{pmatrix} e^{-0.5t} - 2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-2t} \\ \Rightarrow \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} &= \begin{pmatrix} 3e^{-0.5t} \\ 6e^{-0.5t} \end{pmatrix} - \begin{pmatrix} -2e^{-2t} \\ 2e^{-2t} \end{pmatrix} \end{aligned}$$

Therefore,

$$x_1(t) = 3e^{-0.5t} + 2e^{-2t}$$

$$x_2(t) = 6e^{-0.5t} - 2e^{-2t}$$

Sample Exercise-4.1

1. Find the eigenvalues and eigenvectors of the following matrices

a. $\begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$ Ans: $\lambda_1 = -1, \lambda_2 = 5, \mathbf{V}_1 = (-2a, a)^T, \mathbf{V}_2 = (b, b)^T$

b. $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ Ans: $\lambda_1 = 1, \lambda_2 = 3, \mathbf{V}_1 = (-a, a)^T, \mathbf{V}_2 = (b, b)^T$

c. $\begin{pmatrix} 1 & 0 \\ 1 & 3 \end{pmatrix}$ Ans: $\lambda_1 = 1, \lambda_2 = 3, \mathbf{V}_1 = (-2a, a)^T, \mathbf{V}_2 = (0, b)^T$

d. $\begin{pmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$ Ans: $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3$
 $\mathbf{V}_1 = (0, a, 0)^T, \mathbf{V}_2 = (-b, 2b, 2b)^T, \mathbf{V}_3 = (-c, c, c)^T$

e. $\begin{pmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{pmatrix}$ Ans: $\lambda_1 = -2, \lambda_2 = 4, \lambda_3 = 6$
 $\mathbf{V}_1 = (0, a, 0)^T, \mathbf{V}_2 = (b, 0, -b)^T, \mathbf{V}_3 = (c, 0, c)^T$

2. Solve the following system of differential equations using eigenvalue and eigenvector where $\dot{x}_1(t) = \frac{dx_1}{dt}$ and $\dot{x}_2(t) = \frac{dx_2}{dt}$.

a. $\begin{cases} \dot{x}_1(t) = -x_1(t) + 4x_2(t) \\ \dot{x}_2(t) = 3x_1(t) - 2x_2(t) \end{cases}$ with $x_1(0) = 3, x_2(0) = 4$. Ans. $\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = e^{2t} \begin{pmatrix} 4 \\ 3 \end{pmatrix} + e^{-5t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$.

b. $\begin{cases} \dot{x}_1(t) = 5x_1(t) + x_2(t) \\ \dot{x}_2(t) = x_1(t) + 5x_2(t) \end{cases}$ with $x_1(0) = 1, x_2(0) = -3$. Ans. $\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = -e^{6t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 2e^{4t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

c. $\begin{cases} \dot{x}_1(t) = -2x_1 + 3x_2(t) \\ \dot{x}_2(t) = 4x_1(t) - x_2(t) \end{cases}$ with $x_1(0) = 4, x_2(0) = 3$. Ans. $\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = e^{2t} \begin{pmatrix} 3 \\ 4 \end{pmatrix} - e^{5t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

Cayley-Hamilton Theorem:

Every square matrix satisfies its characteristic equation, if the characteristic equation is

$$\lambda^n + a_{n-1}\lambda^{n-1} + a_{n-2}\lambda^{n-2} + \dots + a_1\lambda + a_0I = 0$$

$$\text{Then } A^n + a_{n-1}A^{n-1} + a_{n-2}A^{n-2} + \dots + a_1A + a_0I = 0$$

Example: Verify the Cayley-Hamilton theorem for the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix}$

$$\begin{aligned} \text{Solution: The characteristic matrix of } A \text{ is } A - \lambda I &= \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1-\lambda & 2 & 3 \\ 2 & -1-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{pmatrix} \end{aligned}$$

Therefore, the characteristic equation of the matrix A is $|A - \lambda I| = 0$

$$\begin{aligned} \begin{vmatrix} 1-\lambda & 2 & 3 \\ 2 & -1-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{vmatrix} &= 0 \\ \Rightarrow \lambda^3 - \lambda^2 - 15\lambda - 15 &= 0 \end{aligned}$$

According to Cayley-Hamilton theorem

$$A^3 - A^2 - 15A - 15I = 0$$

$$\text{Now, } A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix}$$

$$A^2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 14 & 3 & 8 \\ 3 & 6 & 6 \\ 8 & 6 & 11 \end{pmatrix}$$

$$A^3 = A^2A = \begin{pmatrix} 14 & 3 & 8 \\ 3 & 6 & 6 \\ 8 & 6 & 11 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 44 & 33 & 53 \\ 33 & 6 & 21 \\ 53 & 21 & 41 \end{pmatrix}$$

$$\therefore A^3 - A^2 - 15A - 15I$$

$$\begin{aligned} &= \begin{pmatrix} 44 & 33 & 53 \\ 33 & 6 & 21 \\ 53 & 21 & 41 \end{pmatrix} - \begin{pmatrix} 14 & 3 & 8 \\ 3 & 6 & 6 \\ 8 & 6 & 11 \end{pmatrix} - 15 \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix} - 15 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0 \end{aligned}$$

$$\therefore A^3 - A^2 - 15A - 15I = 0$$

Hence Cayley-Hamilton theorem is verified.

Example: Using Cayley-Hamilton theorem find the inverse of the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix}$

Solution: The characteristic matrix of A is $A - \lambda I = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$$= \begin{pmatrix} 1-\lambda & 2 & 3 \\ 2 & -1-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{pmatrix}$$

Therefore, the characteristic equation of the matrix A is $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 2 & 3 \\ 2 & -1-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - \lambda^2 - 15\lambda - 15 = 0$$

Now according to the Cayley-Hamilton theorem, we have

$$A^3 - A^2 - 15A - 15I = 0$$

$$\Rightarrow A^2 - A - 15I - 15A^{-1} = 0 \quad [\text{Multiplying by } A^{-1}]$$

$$\Rightarrow A^{-1} = \frac{1}{15}[A^2 - A - 15I]$$

Here,

$$A^2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 14 & 3 & 8 \\ 3 & 6 & 6 \\ 8 & 6 & 11 \end{pmatrix}$$

$$\therefore A^{-1} = \frac{1}{15} \left(\begin{pmatrix} 14 & 3 & 8 \\ 3 & 6 & 6 \\ 8 & 6 & 11 \end{pmatrix} - \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix} - 15 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right)$$

$$= \frac{1}{15} \begin{pmatrix} -2 & 1 & 5 \\ 1 & -8 & 5 \\ 5 & 5 & -5 \end{pmatrix}.$$

Example: Using Cayley-Hamilton theorem find the inverse of the matrix $A = \begin{pmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$

The characteristic matrix of A is $A - \lambda I = \begin{pmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$$= \begin{pmatrix} 1-\lambda & 2 & 2 \\ 3 & 1-\lambda & 0 \\ 1 & 1 & 1-\lambda \end{pmatrix}.$$

Therefore, the characteristic equation of the matrix A is $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 2 & 2 \\ 3 & 1-\lambda & 0 \\ 1 & 1 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - 3\lambda^2 - 5\lambda + 1 = 0$$

Now according to the Cayley –Hamilton theorem, we have

$$A^3 - 3A^2 - 5A + I = 0$$

$$\Rightarrow A^2 - 3A - 5I + A^{-1} = 0$$

$$A^{-1} = 3A + 5I - A^2$$

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$$A^2 = \begin{pmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 9 & 6 & 4 \\ 6 & 7 & 6 \\ 3 & 4 & 3 \end{pmatrix}$$

$$\therefore A^{-1} = 3 \begin{pmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} + 5 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 9 & 6 & 4 \\ 6 & 7 & 6 \\ 3 & 4 & 3 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 2 \\ 3 & 1 & -6 \\ -2 & -1 & 5 \end{pmatrix}$$

Sample Exercise-4.2

State the Cayley-Hamilton theorem. Hence find the inverse of the following matrices using Cayley-Hamilton theorem and verify your result.

a. $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

Ans. $A^{-1} = \frac{1}{2} \begin{pmatrix} -4 & 2 \\ 3 & -1 \end{pmatrix}$

b. $A = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{pmatrix}$

Ans. $A^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{pmatrix}$

c. $A = \begin{pmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{pmatrix}$

Ans. $A^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{pmatrix}$

d. $A = \begin{pmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{pmatrix}$

Ans. $A^{-1} = \begin{pmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$

e. $A = \begin{pmatrix} 1 & -1 & 1 \\ 4 & 1 & 0 \\ 8 & 1 & 1 \end{pmatrix}$

Ans. $A^{-1} = \begin{pmatrix} 1 & 2 & -1 \\ -4 & -7 & 4 \\ -4 & -9 & 5 \end{pmatrix}$

Vector

Vector Spaces

Vectors in $\mathbb{R}^2$:

The set of all ordered pairs of real numbers is called two-dimensional vector space and is denoted by $\mathbb{R}^2$.

$$\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}, \text{ where } \mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$$

Example: $(1, 2); (2, -1); (3, 4); (\sqrt{3}, 5); (0, e) \in \mathbb{R}^2$

Vectors in $\mathbb{R}^3$:

The set of all ordered triplets of real numbers is called three-dimensional vector space and is denoted by $\mathbb{R}^3$

$$\mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}, \text{ where } \mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R}$$

Vectors in $\mathbb{R}^n$:

If n is a positive integer then the set of all ordered n tuples of real numbers is called vector in n -dimensional space and is denoted by $\mathbb{R}^n$ and if $\mathbf{u} \in \mathbb{R}^n$; $\mathbf{u} = (\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n)$, then $\mathbf{u}$ is called a n -dimensional vector in $\mathbb{R}^n$. In Cartesian co-ordinate system a particular n tuples in $\mathbb{R}^n$ is called co-ordinates of a point.

Addition of two vectors in $\mathbb{R}^2$:

If $\mathbf{a} = (a_1, a_2)$ and $\mathbf{b} = (b_1, b_2)$ be two vectors in $\mathbb{R}^2$ then $\mathbf{a} + \mathbf{b} = (a_1 + b_1, a_2 + b_2)$

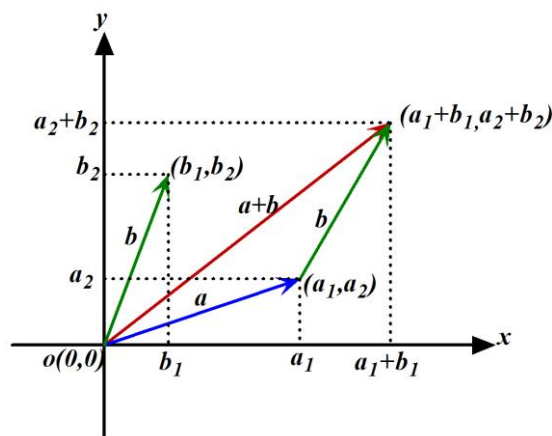


Fig: addition of two vectors.

Subtraction of two vectors in $\mathbb{R}^2$:

If $\mathbf{a} = (a_1, a_2)$ and $\mathbf{b} = (b_1, b_2)$ be two vectors in $\mathbb{R}^2$ then $\mathbf{a} - \mathbf{b} = (a_1 - b_1, a_2 - b_2)$

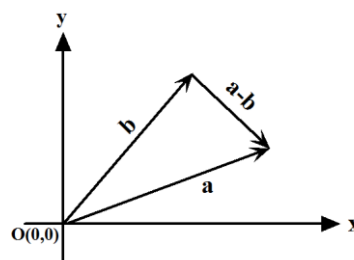


Fig: Subtraction of two vectors.

Scalar multiplication of vectors in $\mathbb{R}^2$:

Let $k \in \mathbb{R}$ be a scalar and $\mathbf{u} \in \mathbb{R}^2$ where $\mathbf{u} = (u_1, u_2)$ is a vector in $\mathbb{R}^2$ then

$$k\mathbf{u} = k(\mathbf{u}_1, \mathbf{u}_2) = (k\mathbf{u}_1, k\mathbf{u}_2)$$

Zero Vector:

The vector whose components are all zero is called the zero vector and is denoted by $\mathbf{0}$ and defined by $\mathbf{0} = (0, 0, \dots, 0)$

Dot or Inner product of two non-zero vectors:

Let $\mathbf{u}(u_1, u_2)$ and $\mathbf{v}(v_1, v_2)$ be two non-zero vectors and θ be the angle between them. The dot product of $\mathbf{u}$ and $\mathbf{v}$ is denoted by $\mathbf{u} \cdot \mathbf{v}$ and defined by $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta = u_1 v_1 + u_2 v_2$.

Parallel vectors in $\mathbb{R}^2$:

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$, then the vectors $\mathbf{u}$ and $\mathbf{v}$ are parallel if $\mathbf{u} = k\mathbf{v}$. If $k > 0$ then they are in the same direction and if $k < 0$ then they are in opposite direction.

Perpendicular vectors in $\mathbb{R}^2$:

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ then $\mathbf{u}$ and $\mathbf{v}$ are said to be perpendicular (or orthogonal) if $\mathbf{u} \cdot \mathbf{v} = 0$.

Example: Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ where $\mathbf{u} = (3, -4)$, $\mathbf{v} = (8, 6)$; $\mathbf{u} \cdot \mathbf{v} = 3(8) + (-4)(6) = 0$.

Hence $\mathbf{u}$ and $\mathbf{v}$ are perpendicular vectors.

Distance between two vectors in $\mathbb{R}^2$:

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ where $\mathbf{u}(u_1, u_2)$ and $\mathbf{v}(v_1, v_2)$, then the distance between $\mathbf{u}$ and $\mathbf{v}$, denoted by $d(\mathbf{u}, \mathbf{v})$ is defined by $d(\mathbf{u}, \mathbf{v}) = \sqrt{(v_1 - u_1)^2 + (v_2 - u_2)^2}$

Example: If $\mathbf{u} = (5, 4)$, $\mathbf{v} = (1, 1)$ then $d(\mathbf{u}, \mathbf{v}) = \sqrt{(5 - 1)^2 + (4 - 1)^2} = \sqrt{16 + 9} = 5$.

Norm or Length in $\mathbb{R}^2$:

Let $\mathbf{u} \in \mathbb{R}^2$, where $\mathbf{u} = (u_1, u_2)$ then the norm or length of $\mathbf{u}$ denoted by $\|\mathbf{u}\|$ and is defined by $\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}} = \sqrt{u_1^2 + u_2^2}$

Example: Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$; where $\mathbf{u} = (1, 0, -1)$, $\mathbf{v} = (2, -3, 1)$ then find

$$(i) 2\mathbf{u} + 3\mathbf{v} \quad (ii) \mathbf{u} \cdot \mathbf{v} \quad (iii) \|\mathbf{v}\| \quad (iv) \|\mathbf{u} - \mathbf{v}\| \quad (v) d(\mathbf{u}, \mathbf{v})$$

Solution:

$$(i) 2\mathbf{u} + 3\mathbf{v} = 2(1, 0, -1) + 3(2, -3, 1) = (2, 0, -2) + (6, -9, 3) = (8, -9, 1)$$

$$(ii) \mathbf{u} \cdot \mathbf{v} = 1(2) + 0(-3) + (-1)(1) = 2 - 1 = 1$$

$$(iii) \|\mathbf{v}\| = \sqrt{4 + 9 + 1} = \sqrt{14}$$

$$(iv) \mathbf{u} - \mathbf{v} = (1 - 2, 0 + 3, -1 - 1) = (-1, 3, -2)$$

$$\therefore \|\mathbf{u} - \mathbf{v}\| = \sqrt{(-1)^2 + 3^2 + (-2)^2} = \sqrt{14}$$

$$(v) d(\mathbf{u}, \mathbf{v}) = \sqrt{(1 - 2)^2 + (0 + 3)^2 + (-1 - 1)^2} = \sqrt{14}.$$

Vector Spaces:

Let F be a field of scalars and V be a non-empty set of vectors. If V satisfies the following axioms of vector addition and scalar multiplication, then V is called vector space over F .

In vector addition:

$$A_1: \mathbf{u}, \mathbf{v} \in V \Rightarrow (\mathbf{u} + \mathbf{v}) \in V$$

$$A_2: \mathbf{u}, \mathbf{v}, \mathbf{w} \in V \Rightarrow (\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$$

$$A_3: \mathbf{u}, \mathbf{0} \in V \Rightarrow (\mathbf{u} + \mathbf{0}) = (\mathbf{0} + \mathbf{u}) = \mathbf{u}, \quad \text{here } \mathbf{0} \text{ is the zero vector}$$

$$A_4: \mathbf{u}, \mathbf{v} \in V \Rightarrow \mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

$$A_5: \mathbf{u} \in V \Rightarrow -\mathbf{u} \in V \Rightarrow \mathbf{u} + (-\mathbf{u}) = (-\mathbf{u}) + \mathbf{u} = \mathbf{0}$$

In scalar multiplication:

$$M_1: a \in F \text{ and } \mathbf{u} \in V \Rightarrow a\mathbf{u} \in V$$

$$M_2: a \in F \text{ and } \mathbf{u}, \mathbf{v} \in V \Rightarrow a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$$

$$M_3: a, b \in F \text{ and } \mathbf{u} \in V \Rightarrow (a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$$

$$M_4: a, b \in F \text{ and } \mathbf{u} \in V \Rightarrow (ab)\mathbf{u} = a(b\mathbf{u})$$

$$M_5: 1 \in F \Rightarrow 1.\mathbf{u} = \mathbf{u}.1 = \mathbf{u}; \quad \mathbf{u} \in V$$

Subspace:

Let W be a non empty subset of a vector space V over the field F . We call W a subspace of V if and only if W is a vector space over the field F under the laws of vector addition and scalar multiplication defined on V , or equivalently, W is a subspace of V wherever $\mathbf{w}_1, \mathbf{w}_2 \in W$ and $\alpha, \beta \in F$ implies that $\alpha\mathbf{w}_1 + \beta\mathbf{w}_2 \in W$.

Example: Show that $S = \{(a, b, c) \mid a, b, c \in \mathbb{R} \text{ and } a + b + c = 0\}$ is a subspace of the vector space $\mathbb{R}^3$.

Solution: Here $\mathbf{0} \in \mathbb{R}^3$, $\mathbf{0} = (0, 0, 0) \in S \quad \therefore 0 + 0 + 0 = 0$

Hence S is non empty set. Again, let $\mathbf{u} = (a_1, b_1, c_1) \in S$ and $\mathbf{v} = (a_2, b_2, c_2) \in S$

$$a_1 + b_1 + c_1 = 0 \quad \text{and} \quad a_2 + b_2 + c_2 = 0$$

For any scalars α, β we get $\alpha\mathbf{u} + \beta\mathbf{v} = \alpha(a_1, b_1, c_1) + \beta(a_2, b_2, c_2)$

$$= (\alpha a_1 + \beta a_2, \alpha b_1 + \beta b_2, \alpha c_1 + \beta c_2)$$

$$\text{Now, } \alpha a_1 + \beta a_2 + \alpha b_1 + \beta b_2 + \alpha c_1 + \beta c_2$$

$$= \alpha(a_1 + b_1 + c_1) + \beta(a_2 + b_2 + c_2) = \alpha.0 + \beta.0 = 0$$

$\therefore \alpha\mathbf{u} + \beta\mathbf{v} \in S$. Hence S is a subspace of $\mathbb{R}^3$.

Linear combination, Dependency and Independency of Vector**Linear combination:**

Let $V(F)$ be a vector space, where $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \dots \mathbf{v}_n \in V$ and $\alpha_1, \alpha_2, \alpha_3 \dots \alpha_n \in F$. If $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3 + \dots + \alpha_n\mathbf{v}_n = \mathbf{u} \in V$, then $\mathbf{u}$ is called linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \dots \mathbf{v}_n$.

Linear dependency of vectors:

Let $V(F)$ be a vector space, where $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \dots \mathbf{v}_n \in V$ and $\alpha_1, \alpha_2, \alpha_3 \dots \alpha_n \in F$. If $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 \mathbf{v}_3 + \dots + \alpha_n \mathbf{v}_n = \mathbf{0}$, and at least one of the element of the set $\{\alpha_1, \alpha_2, \alpha_3 \dots \alpha_n\}$ is not zero, then the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \dots \mathbf{v}_n$ are linearly dependent.

Linear independency of vectors:

Let $V(F)$ be a vector space and $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots \mathbf{v}_n \in V$ and $\alpha_1, \alpha_2, \alpha_3, \dots \alpha_n \in F$. If $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 \mathbf{v}_3 + \dots + \alpha_n \mathbf{v}_n = \mathbf{0}$, and all of the elements of the set $\{\alpha_1, \alpha_2, \alpha_3 \dots \alpha_n\}$ are zero except the trivial solution ($\alpha_1 = \alpha_2 = \dots = \alpha_n = 0$), then the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \dots \mathbf{v}_n$ are linearly independent.

Example: Write the vector $\mathbf{u} = (1, -2, 5)$ as a linear combination of the vectors

$$\mathbf{u}_1 = (1, 1, 1), \quad \mathbf{u}_2 = (1, 2, 3) \text{ and } \mathbf{u}_3 = (2, -1, 1).$$

Solution: Let $\alpha_1 \mathbf{u}_1 + \alpha_2 \mathbf{u}_2 + \alpha_3 \mathbf{u}_3 = \mathbf{u}$; where $\alpha_1, \alpha_2, \alpha_3$ are scalars.

$$\alpha_1(1, 1, 1) + \alpha_2(1, 2, 3) + \alpha_3(2, -1, 1) = (1, -2, 5)$$

$$(\alpha_1 + \alpha_2 + 2\alpha_3, \alpha_1 + 2\alpha_2 - \alpha_3, \alpha_1 + 3\alpha_2 + \alpha_3) = (1, -2, 5)$$

Equating corresponding components

$$\alpha_1 + \alpha_2 + 2\alpha_3 = 1$$

$$\alpha_1 + 2\alpha_2 - \alpha_3 = -2$$

$$\alpha_1 + 3\alpha_2 + \alpha_3 = 5$$

The augmented matrix of the above system is

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 1 & 2 & -1 & -2 \\ 1 & 3 & 1 & 5 \end{array} \right)$$

Reducing the system to REF by the elementary row operations

$$\begin{array}{lcl} r_2 \rightarrow r_2 - r_1 & \sim & \left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & -3 & -3 \\ 0 & 2 & -1 & 4 \end{array} \right) \\ r_3 \rightarrow r_3 - r_1 & & \\ r_3 \rightarrow r_3 - 2r_2 & \sim & \left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & -3 & -3 \\ 0 & 0 & 5 & 10 \end{array} \right) \\ r_3 \rightarrow \frac{1}{5}r_3 & \sim & \left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & -3 & -3 \\ 0 & 0 & 1 & 2 \end{array} \right) \end{array}$$

So, system of linear equations becomes,

$$\alpha_1 + \alpha_2 + 2\alpha_3 = 1$$

$$\alpha_2 - 3\alpha_3 = -3$$

$$\alpha_3 = 2$$

Solving, we get $\alpha_1 = -6, \alpha_2 = 3$ and $\alpha_3 = 2$.

$\therefore -6\mathbf{u}_1 + 3\mathbf{u}_2 + 2\mathbf{u}_3 = \mathbf{u}$. Hence $\mathbf{u}$ is a linear combination of the vectors $\mathbf{u}_1, \mathbf{u}_2$ and $\mathbf{u}_3$.

Example: Test whether the vectors $\mathbf{u}_1 = (1, 0, 1)$, $\mathbf{u}_2 = (-3, 2, 6)$ & $\mathbf{u}_3 = (4, 5, -2)$ are linearly dependent or independent.

Solution: Let $\alpha_1 \mathbf{u}_1 + \alpha_2 \mathbf{u}_2 + \alpha_3 \mathbf{u}_3 = \mathbf{0}$; where $\alpha_1, \alpha_2, \alpha_3$ are scalars.

$$\alpha_1(1, 0, 1) + \alpha_2(-3, 2, 6) + \alpha_3(4, 5, -2) = (0, 0, 0)$$

$$(\alpha_1 - 3\alpha_2 + 4\alpha_3, 0 + 2\alpha_2 + 5\alpha_3, \alpha_1 + 6\alpha_2 - 2\alpha_3) = (0, 0, 0)$$

We can write from above equation

$$\alpha_1 - 3\alpha_2 + 4\alpha_3 = 0$$

$$0 + 2\alpha_2 + 5\alpha_3 = 0$$

$$\alpha_1 + 6\alpha_2 - 2\alpha_3 = 0$$

The augmented matrix of the above system is

$$\left(\begin{array}{ccc|c} 1 & -3 & 4 & 0 \\ 0 & 2 & 5 & 0 \\ 1 & 6 & -2 & 0 \end{array} \right)$$

Reducing the system to REF by the elementary row operations

$$\begin{aligned} r_3 \rightarrow r_3 - r_1 & \sim \left(\begin{array}{ccc|c} 1 & -3 & 4 & 0 \\ 0 & 2 & 5 & 0 \\ 0 & 9 & -6 & 0 \end{array} \right) \\ r_3 \rightarrow 2r_3 - 9r_2 & \sim \left(\begin{array}{ccc|c} 1 & -3 & 4 & 0 \\ 0 & 2 & 5 & 0 \\ 0 & 0 & -57 & 0 \end{array} \right) \\ r_3 \rightarrow \frac{1}{-57}r_3 & \sim \left(\begin{array}{ccc|c} 1 & -3 & 4 & 0 \\ 0 & 2 & 5 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right) \end{aligned}$$

So, system of linear equations becomes,

$$\alpha_1 - 3\alpha_2 + 4\alpha_3 = 0$$

$$2\alpha_2 + 5\alpha_3 = 0$$

$$\alpha_3 = 0$$

Solving the above linear system, we get $\alpha_1 = 0$, $\alpha_2 = 0$ and $\alpha_3 = 0$.

Hence the vectors $\mathbf{u}_1, \mathbf{u}_2$ and $\mathbf{u}_3$ are linearly independent.

Sample Exercise-4.3

Write the vector $\mathbf{u}$ as a linear combination of the vectors $\mathbf{u}_1$, $\mathbf{u}_2$ and $\mathbf{u}_3$, where

1. $\mathbf{u} = (5, -1, 9)$, $\mathbf{u}_1 = (1, 2, 1)$, $\mathbf{u}_2 = (2, 9, 0)$, $\mathbf{u}_3 = (3, 3, 4)$. **Ans:** $(1, -1, 2)$
2. $\mathbf{u} = (2, -1, 3)$, $\mathbf{u}_1 = (1, 0, 0)$, $\mathbf{u}_2 = (2, 2, 0)$, $\mathbf{u}_3 = (3, 3, 3)$. **Ans:** $(3, -2, 1)$
3. $\mathbf{u} = (4, 2, 1, 0)$, $\mathbf{u}_1 = (6, -1, 2, 1)$, $\mathbf{u}_2 = (1, 7, -3, -2)$, $\mathbf{u}_3 = (3, 1, 0, 0)$,
 $\mathbf{u}_4 = (3, 3, -2, -1)$. **Ans:** $(2, 1, -3, 0)$.

Test whether the following vectors are linearly independent or dependent.

4. $\mathbf{u}_1 = (1, -2, 3)$, $\mathbf{u}_2 = (5, 6, -1)$, $\mathbf{u}_3 = (3, 2, 1)$. **Ans:** Linearly dependent.
5. $\mathbf{u}_1 = (1, 2, 2, -1)$, $\mathbf{u}_2 = (4, 9, 9, -4)$, $\mathbf{u}_3 = (5, 8, 9, -5)$. **Ans:** Linearly independent.
6. $\mathbf{u}_1 = (-3, 0, 4)$, $\mathbf{u}_2 = (5, -1, 2)$, $\mathbf{u}_3 = (1, 1, 3)$. **Ans:** Linearly independent.
7. $\mathbf{u}_1 = (2, 2, 1)$, $\mathbf{u}_2 = (-1, -1, 0)$, $\mathbf{u}_3 = (3, 3, 4)$. **Ans:** Linearly dependent.
8. $\mathbf{u}_1 = (-2, 0, 1)$, $\mathbf{u}_2 = (3, 2, 5)$, $\mathbf{u}_3 = (6, -1, 1)$, $\mathbf{u}_4 = (7, 0, -2)$.
Ans: Linearly dependent.